

Password Security Assessment: Dictionary and Brute-Force Attack Analysis

EXECUTIVE SUMMARY

This project evaluates password security through controlled dictionary and brute-force attack simulations using **John the Ripper**. Multiple encrypted ZIP files protected with passwords of varying complexity—including letters, numbers, alphanumeric combinations, and special characters—were assessed to examine their resistance against common password-cracking techniques. The project compares the effectiveness of dictionary attacks using standard wordlists with brute-force attacks that systematically generate password combinations.

An advanced password recovery scenario was also performed by developing a custom Python script to decode a Caesar cipher and generate a tailored wordlist, demonstrating how attackers can adapt their techniques when conventional password lists fail. The results highlight the significant impact of password length, complexity, and unpredictability on resistance to password-cracking attacks while reinforcing the importance of secure password policies, multi-factor authentication (MFA), password hashing, and user awareness. This project provides practical insight into password auditing methodologies and emphasizes defensive strategies for strengthening authentication security against credential-based attacks.

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INTRODUCTION

The unwanted access to computer systems, files, and accounts is prevented by the use of passwords. Nevertheless, many individuals have weak passwords that can be easily deciphered or broken by the hackers. In order to crack passwords and gain unauthorized access to systems, hackers often use automated tools and comprehensive passwords (Conti et al., 2018). The purpose of this experiment was to identify the ways in which a dictionary and brute-force attacks can be employed to crack passwords.

In this lab, the encrypted passwords of files in ZIP format were tried to be retrieved by attempting to recover them with the aid of the program John the Ripper. The four types of the tested passwords were only letters, only numbers, letters and numbers, and letters and numbers with a special character. Each type was done using one weak and one strong password. The results proved that strong passwords are more challenging to crack as compared to weak passwords. This proves the importance of having a strong password where there is enhanced security.

TOOLS AND RESOURCES USED

The following tools and resources were used in this experiment :

- 1) **John the Ripper** – A password cracking tool is used to implement dictionary and brute-force attacks (Openwall, 2024).
- 2) **RockYou Wordlist** – The leaked passwords were a real world data breach of millions of passwords contained in a popular password list (Florocio and Herley, 2007).
- 3) **Large Password Wordlist (28.36 million passwords)** – Used to increase the chances that weak passwords will be found.
- 4) **ZIP Files** – ZIP files had to be prepared and encrypted using different passwords for testing.

John the Ripper was applied to try to obtain the passwords in the ZIP files using both dictionary and bruteforce attacks.

METHODOLOGY

This experiment came up with four types of passwords used to study the relationship between password security and password strength. Each type of password was done with two passwords; one weak password and one strong password.

Each password was tested using two attack methods:

- 1) Dictionary attack (wordlist attack)
- 2) Brute-force attack.

The two attacks made use of John the Ripper. This enabled us to make a comparison between cracking weak and strong passwords easily.

➤ Password Categories Tested :

1) Passwords with Only Letters

✚ Weak password : pass

✚ Strong password : pazzrtfg

2) Passwords with Only Numbers

✚ Weak password : 1234

✚ Strong password : 67384567

3) Passwords with Letters and Numbers

✚ Weak password : pass67

✚ Strong password : qwedf44535

4) Passwords with Letters, Numbers and Special Characters

✚ Weak password : Pass671!

✚ Strong password : qweDf44535*

ATTACK SIMULATION

Two methods of password cracking were performed in this experiment using John the Ripper to access passwords of files in encrypted ZIP files.

➤ Dictionary Attack

A dictionary attack also compares encrypted passwords to wordlist passwords until the correct password is found (Stallings, 2017). In the given experiment, John the Ripper tried to break the passwords in ZIP files with the help of a massive passwords list and RockYou wordlist.

These wordlists contain millions of popular passwords. When the users choose weak or popular passwords such as simple words or predictable combinations, the dictionary attacks are highly effective.

➤ Brute-Force Attack

A brute force attack will check all the combinations of characters trying to guess the password (Conti et al., 2018). This method does not use a wordlist, which is unlike dictionary attacks.

Instead, until the correct password is found, the tool will generate combinations of letters, numbers and symbols. However, brute-force attacks may require an extremely long time, especially when the password is longer, or its characters are more complex.

➤ Advanced Scenario

I were sent two zip files, one of which was called key.txt and the other one main.txt. The hashes were first of course extracted with John the Ripper. After that, I were able to crack a password by executing our own wordlist on the key.txt hash. However, the password failed to open the key.zip file.

I then learned that the actual password of key.zip was: ccipher implying that it was a Caesar cipher. To solve this, I created a Python script that cycles through 26 letter shifts and outputs the results into a new file. This file was used to develop a new custom wordlist.

I finally managed to find the correct password by matching this new wordlist with main.zip.

RESULTS AND FINDINGS

The experiment showed the importance of using strong passwords to protect files and accounts. Passwords of strength proved to be very hard to break as compared to weak passwords.

➤ Weak Passwords

The weak passwords of the experiment were :

- ‡ pass
- ‡ 1234
- ‡ pass67
- ‡ Pass671!

These passwords could easily be broken by both brute force and dictionary attacks. Some of them had high frequency and thus were easily detected in the wordlist. Also, they are short and, therefore, the brute-force method can guess them more easily.

➤ Strong Passwords

The strong passwords of the experiment were:

- ‡ pazzrtfg
- ‡ 67384567
- ‡ qwedf44535
- ‡ qweDf44535*

These passwords were more difficult to determine. The dictionary attack did not often find the password in the wordlist. The process was time-consuming since the passwords were longer and included more complex characters during the time when brute-force was applied. This shows that the longer and more complex a password is, the greater a heightened level of security can be achieved.

Password Type	Strength	Password	Dictionary Attack	Brute-Force Attack
Only letters	Weak	pass	Cracked	Cracked
Only letters	Strong	pazzrtfg	Not found	Took too long
Only numbers	Weak	1234	Cracked	Cracked
Only numbers	Strong	67384567	Not found	Took a long time
Letters + numbers	Weak	pass67	Not found	Cracked
Letters + numbers	Strong	qwedf44535	Not found	Took too long
Letters + numbers + special	Weak	Pass671!	Not found	Took too long
Letters + numbers + special	Strong	qweDf44535*	Not found	Took too long

TABLE 1: SUMMARY OF DICTIONARY AND BRUTE-FORCE ATTACK RESULTS

ANALYSIS OF PASSWORD VULNERABILITIES

The experiment provided some of the most common password weaknesses.

1) Short Password Length

Due to the many different combinations of short passwords such as 1234, they are extremely easy to crack.

2) Use of Common Words

Common passwords such as pass are most likely to be used and include in many password wordlists. Owing to this, they can be easily attacked by dictionary attacks.

3) Predictable Patterns

Even passwords that are composed of simple words such as pass67 with the addition of numbers are relatively easy to crack.

4) Lack of Character Variety

Passwords that consist of letters alone or numbers alone are easily deciphered as compared to those using special characters and capital letters.

SECURITY RECOMMENDATIONS

The findings of the experiment show that there are a number of security measures that can be employed in order to strengthen.

1) Strong Password Requirements

Organizations should require the use of strong passwords. It should be a powerful password with at least 12 characters. Capital and lower case letters, digits and special characters. These are tough passwords with which brute-force or dictionary attacks are hard to crack.

2) Secure Password Storage

They will not be stored as plain text passwords; they will be encrypted using secure hash functions, e.g. bcrypt, Argon2, or PBKDF2 (NIST, 2023). The methods make passwords much harder to access by attackers.

3) Account Lockout Mechanism

Systems should limit the number of failed attempts to log-in. Examples would be that an account can be temporarily locked down after 5 unsuccessful attempts or a delay may occur between log-in attempts. This reduces the probability of brutality attacks.

4) Multi-Factor Authentication (MFA)

Multi-factor authentication also provides an extra layer of protection to the user accounts (NIST, 2023). Users have to authenticate themselves by another means, such as an SMS code, an authentication app, or by using biometrics. This ensures that accounts are secured even where password is broken.

CONCLUSION

In this experiment, the authors were able to identify passwords with password cracking tools such as John the Ripper through dictionary attacks and brute-force attacks. The results showed that the passwords were easily broken because they were simple or were in the list of popular passwords. On the contrary, strong passwords were far harder to decrypt, as they contained a lot of characters, comprised of special characters, numbers, and letters.

In some cases the password was not recoverable due to the excessive long time of the brute-force attack. On the whole, this experiment demonstrates that it is critical to use safe passwords to protect systems and accounts. Organizations should also employ other security besides ensuring that users use strong passwords features such as multi factor authentication, password lockout policies and password hashing.

REFERENCES

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Stallings, W., 2017. *Cryptography and network security: Principles and practice*. 7th ed. Harlow: Pearson.

➤ Appendix A: Password Cracking

```
(kali@kali)-[~]
$ john --wordlist=/home/kali/Desktop/linuxwz.txt /home/kali/Desktop/hashz.txt
Using default input encoding: UTF-8
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])
Cost 1 (HMAC size) is 564 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:04 0.98% (ETA: 09:18:44) 0g/s 67760p/s 67760c/s 67760C/s P!ecrvtpjsolfo
3htq..Fischen69
0g 0:00:00:07 1.72% (ETA: 09:18:44) 0g/s 68654p/s 68654c/s 68654C/s apusyzeg..Bett
enluder
0g 0:00:00:10 2.45% (ETA: 09:18:44) 0g/s 68946p/s 68946c/s 68946C/s SpeckrinG..Br0
gan123
pass (steps.zip/steps)
1g 0:00:01:14 DONE (2026-03-09 09:13) 0.01334g/s 64104p/s 64104c/s 64104C/s Epa317
537..traviesa67
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

(kali@kali)-[~]
$
```

FIGURE 1:THIS IS THE FIRST ONE WITH PASSWORD, WORDLIST, PASSWORD “PASS”

```
kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
$ john --incremental /home/kali/Desktop/hashz.txt
Using default input encoding: UTF-8
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])
Cost 1 (HMAC size) is 564 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
pass (steps.zip/steps)
1g 0:00:00:01 DONE (2026-03-09 09:16) 0.7751g/s 41277p/s 41277c/s 41277C/s breads.
.147488
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

(kali@kali)-[~]
$
```

FIGURE 2 : THIS IS THE EASY PASSWORD, ONLY LETTERS, BRUTEFORCE, PASSWORD “PASS”

```
(kali@kali)-[~]
$ john --wordlist=/home/kali/Desktop/linuxwz.txt /home/kali/Desktop/hashz.txt
Using default input encoding: UTF-8
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])
Cost 1 (HMAC size) is 564 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:02 0.26% (ETA: 09:33:02) 0g/s 32386p/s 32386c/s 32386C/s Kolli-1974..Mb
s3162001
0g 0:00:00:18 2.12% (ETA: 09:34:22) 0g/s 33057p/s 33057c/s 33057C/s Fortschritt49.
.Bs8ill0yAXDeDQ==
0g 0:00:00:32 3.77% (ETA: 09:34:22) 0g/s 33135p/s 33135c/s 33135C/s fjgdet5t..fuck
er1369
0g 0:00:01:14 8.65% (ETA: 09:34:29) 0g/s 33046p/s 33046c/s 33046C/s Daboroking123.
.Jaser1234
0g 0:00:01:51 12.96% (ETA: 09:34:30) 0g/s 33045p/s 33045c/s 33045C/s w19g_11e57..D
etersygt123123
0g 0:00:01:55 13.43% (ETA: 09:34:30) 0g/s 33065p/s 33065c/s 33065C/s v37M6mqI6swz.
.BDCULrSD!!
0g 0:00:01:56 13.55% (ETA: 09:34:30) 0g/s 33062p/s 33062c/s 33062C/s tiqiwuhy..Bon
nie99!!
0g 0:00:01:58 13.78% (ETA: 09:34:30) 0g/s 33057p/s 33057c/s 33057C/s sakycumi..Lau
raaa1212123
0g 0:00:01:59 13.89% (ETA: 09:34:30) 0g/s 33055p/s 33055c/s 33055C/s als81ter..Die
anne97
0g 0:00:02:47 19.51% (ETA: 09:34:29) 0g/s 33087p/s 33087c/s 33087C/s karimi01322..
205975125972832
0g 0:00:03:27 24.15% (ETA: 09:34:31) 0g/s 33055p/s 33055c/s 33055C/s 13042001Je..k
g1999
0g 0:00:03:47 26.50% (ETA: 09:34:30) 0g/s 33072p/s 33072c/s 33072C/s mausdorn..nan
yledu
0g 0:00:04:15 29.75% (ETA: 09:34:31) 0g/s 33057p/s 33057c/s 33057C/s asti_453..a01
S2Db
0g 0:00:04:18 30.10% (ETA: 09:34:31) 0g/s 33058p/s 33058c/s 33058C/s 912070039..ba
risA11
0g 0:00:09:23 70.13% (ETA: 09:33:36) 0g/s 35319p/s 35319c/s 35319C/s simmel8100..P
ubg3456@
0g 0:00:09:34 72.76% (ETA: 09:33:22) 0g/s 35937p/s 35937c/s 35937C/s Demence1..pou
polo
0g 0:00:10:44 89.44% (ETA: 09:32:14) 0g/s 39380p/s 39380c/s 39380C/s Michael88!!..
ladyria616
unlink: /home/kali/.john/john.rec: No such file or directory

(kali@kali)-[~]
$ john --show /home/kali/Desktop/hashz.txt
0 password hashes cracked, 1 left

(kali@kali)-[~]
$
```

FIGURE 3:THIS IS ONLY LETTERS, HARD PASSWORD, WITH WORDLIST, PASSWORD
“PAZZRTEFG”

```
(kali@kali)-[~]
$ john --incremental /home/kali/Desktop/hashz.txt
Using default input encoding: UTF-8
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])
Cost 1 (HMAC size) is 564 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:15 0g/s 62134p/s 62134c/s 62134C/s smingi..smemmi
0g 0:00:04:20 0g/s 67429p/s 67429c/s 67429C/s rhs44..hhcj2
0g 0:00:08:02 0g/s 67571p/s 67571c/s 67571C/s kktrr5..km0uns
0g 0:00:09:32 0g/s 67638p/s 67638c/s 67638C/s llyriz..lhuiez
0g 0:00:11:23 0g/s 67661p/s 67661c/s 67661C/s buepjj..bim1t4
0g 0:00:11:39 0g/s 67674p/s 67674c/s 67674C/s detag..dwg2e
0g 0:00:13:15 0g/s 67586p/s 67586c/s 67586C/s dhasgit..dhssiky
0g 0:00:14:35 0g/s 67613p/s 67613c/s 67613C/s jmm2hbb..jmd41ro
0g 0:00:20:06 0g/s 67685p/s 67685c/s 67685C/s ll119072..ll190571
0g 0:00:20:07 0g/s 67684p/s 67684c/s 67684C/s 148lac..15093d
0g 0:00:22:05 0g/s 67651p/s 67651c/s 67651C/s 2bn19r..2bny82
0g 0:00:22:10 0g/s 67653p/s 67653c/s 67653C/s pez054..pujrej
Session aborted
```

FIGURE 4 : THIS IS ONLY LETTERS, HARD PASSWORD, WITH BRUTEFORCE, PASSWORD
“PAZZRTEFG”


```
(kali@kali)-[~]
$ john --wordlist=/home/kali/Desktop/linuxwz.txt /home/kali/Desktop/hashz.txt
Using default input encoding: UTF-8
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])
Cost 1 (HMAC size) is 564 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:04 0.95% (ETA: 10:04:47) 0g/s 66394p/s 66394c/s 66394C/s fizifyta..Sara
hh91
1234 (steps.zip/steps)
1g 0:00:01:57 DONE (2026-03-09 09:59) 0.008542g/s 68161p/s 68161c/s 68161C/s Stig1
989..bicijyce
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

(kali@kali)-[~]
$
```

FIGURE 5 : THIS IS THE EASY PASSWORD, ONLY NUMBERS, WORDLIST, PASSWORD
"1234"

```
(kali@kali)-[~]
$ john --incremental /home/kali/Desktop/hashz.txt
Using default input encoding: UTF-8
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])
Cost 1 (HMAC size) is 564 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
1234 (steps.zip/steps)
1g 0:00:00:00 DONE (2026-03-09 10:01) 3.333g/s 13653p/s 13653c/s 13653C/s 123456..
025246
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

(kali@kali)-[~]
$
```

FIGURE 6 : THIS IS THE EASY PASSWORD, ONLY NUMBERS, BRUTEFORCE, PASSWORD
"1234"

```
(kali@kali)-[~]  
$ john --wordlist=/home/kali/Desktop/linuxwz.txt /home/kali/Desktop/hashz.txt  
Using default input encoding: UTF-8  
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])  
Cost 1 (HMAC size) is 564 for all loaded hashes  
Will run 2 OpenMP threads  
Press 'q' or Ctrl-C to abort, almost any other key for status  
0g 0:00:00:04 0.95% (ETA: 10:10:05) 0g/s 66064p/s 66064c/s 66064C/s fizifyta..Sara  
hh91  
0g 0:00:06:57 DONE (2026-03-09 10:10) 0g/s 67989p/s 67989c/s 67989C/s Albotron..Eg  
or2003  
Session completed.
```

FIGURE 7 : THIS IS THE HARD PASSWORD, ONLY NUMBERS, WORDLIST, PASSWORD
“67384567”

```
(kali@kali)-[~]  
$ john --incremental /home/kali/Desktop/hashz.txt  
Using default input encoding: UTF-8  
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])  
Cost 1 (HMAC size) is 564 for all loaded hashes  
Will run 2 OpenMP threads  
Press 'q' or Ctrl-C to abort, almost any other key for status  
0g 0:00:05:18 0g/s 67425p/s 67425c/s 67425C/s nnkhy..gwzde  
0g 0:00:05:22 0g/s 67439p/s 67439c/s 67439C/s aksh04..ak1264  
Session aborted
```

FIGURE 8 : THIS IS THE HARD PASSWORD, ONLY NUMBERS, BRUTEFORCE, PASSWORD
“67384567”


```

(kali@kali)-[~]
$ john --wordlist=/home/kali/Desktop/linuxwz.txt /home/kali/Desktop/hashz.txt
Using default input encoding: UTF-8
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])
Cost 1 (HMAC size) is 564 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:17 4.13% (ETA: 10:33:02) 0g/s 68426p/s 68426c/s 68426C/s Jensey... Shrif
mayar
0g 0:00:00:19 4.61% (ETA: 10:33:02) 0g/s 68482p/s 68482c/s 68482C/s zemydumy.. szyb
kienrike
0g 0:00:05:30 79.22% (ETA: 10:33:06) 0g/s 68063p/s 68063c/s 68063C/s Cokoladica..C
ooler500
0g 0:00:06:57 DONE (2026-03-09 10:33) 0g/s 67965p/s 67965c/s 67965C/s Albotron..Eg
or2003
Session completed.

```

FIGURE 9 : THIS IS THE EASY PASSWORD, MIX OF NUMBER AND LETTERS, WORDLIST, PASSWORD “PASS67”

```

(kali@kali)-[~]
$ john --incremental /home/kali/Desktop/hashz.txt
Using default input encoding: UTF-8
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])
Cost 1 (HMAC size) is 564 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:34 0g/s 66821p/s 66821c/s 66821C/s mayaly7..meram15
0g 0:00:00:36 0g/s 66903p/s 66903c/s 66903C/s bbbhui..johums
0g 0:00:00:37 0g/s 66920p/s 66920c/s 66920C/s dyeeta..sammeira
0g 0:00:00:38 0g/s 66956p/s 66956c/s 66956C/s litchan..leo1989
0g 0:00:01:50 0g/s 67838p/s 67838c/s 67838C/s 2soshs..lolsch
0g 0:00:02:56 0g/s 68049p/s 68049c/s 68049C/s djkllok..dhyz8m
0g 0:00:03:01 0g/s 68040p/s 68040c/s 68040C/s lhcik..dw6ch
0g 0:00:03:56 0g/s 68136p/s 68136c/s 68136C/s bobilier..bostor13
0g 0:00:03:58 0g/s 68140p/s 68140c/s 68140C/s bmxcg..ccasj
0g 0:00:03:59 0g/s 68138p/s 68138c/s 68138C/s motorai..meg1217
pass67 (steps.zip/steps)
1g 0:00:04:13 DONE (2026-03-09 10:46) 0.003940g/s 68121p/s 68121c/s 68121C/s dc
6..payrke
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

```

FIGURE 10 : THIS IS THE EASY PASSWORD, MIX OF NUMBER AND LETTERS, BRUTEFORCE, PASSWORD “PASS67”

```
(kali@kali)-[~]
$ john --wordlist=/home/kali/Desktop/linuxwz.txt /home/kali/Desktop/hashz.txt
Using default input encoding: UTF-8
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])
Cost 1 (HMAC size) is 564 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:03 0.71% (ETA: 10:54:41) 0g/s 65101p/s 65101c/s 65101C/s schalala..harr
ythepagan
0g 0:00:02:12 31.60% (ETA: 10:54:35) 0g/s 67852p/s 67852c/s 67852C/s Ii8irc..orfqg
r!8gbgdaiwKbw
0g 0:00:04:01 57.52% (ETA: 10:54:36) 0g/s 67666p/s 67666c/s 67666C/s 899James..Ket
chum
0g 0:00:06:24 91.65% (ETA: 10:54:36) 0g/s 67678p/s 67678c/s 67678C/s Cadsys99..lol
kindlol
0g 0:00:06:58 DONE (2026-03-09 10:54) 0g/s 67747p/s 67747c/s 67747C/s Albotron..Eg
or2003
Session completed.

(kali@kali)-[~]
$
```

FIGURE 11 : THIS IS THE HARD PASSWORD, MIX OF NUMBER AND LETTERS, WORDLIST, PASSWORD “QWEDF44535”

```
kali@kali: ~
Session Actions Edit View Help
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:05 0g/s 58514p/s 58514c/s 58514C/s casmy30..benicer
Session aborted

(kali@kali)-[~]
$ john --incremental /home/kali/Desktop/hashz.txt
Using default input encoding: UTF-8
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])
Cost 1 (HMAC size) is 0 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:02 0g/s 51697p/s 51697c/s 51697C/s joybb1..juigee
0g 0:00:02:28 0g/s 68506p/s 68506c/s 68506C/s sed43l..sospr5
0g 0:00:02:33 0g/s 68511p/s 68511c/s 68511C/s pscggi..pjcs43
0g 0:00:02:34 0g/s 68514p/s 68514c/s 68514C/s kmmue2..faisi
0g 0:00:02:37 0g/s 68492p/s 68492c/s 68492C/s abuts25..aputesa
0g 0:00:02:38 0g/s 68495p/s 68495c/s 68495C/s jh19640..jmrियो
0g 0:00:02:39 0g/s 68494p/s 68494c/s 68494C/s pikumel..puca125
0g 0:00:02:40 0g/s 68492p/s 68492c/s 68492C/s singy5..suzo91
0g 0:00:02:43 0g/s 68496p/s 68496c/s 68496C/s crebzb..cemps7
0g 0:00:02:44 0g/s 68499p/s 68499c/s 68499C/s liakpm..lebrl5
0g 0:00:02:45 0g/s 68498p/s 68498c/s 68498C/s pigapt..peojrr
0g 0:00:02:46 0g/s 68496p/s 68496c/s 68496C/s kiwah1..kokkss
0g 0:00:02:47 0g/s 68491p/s 68491c/s 68491C/s munisong..mureme21
0g 0:00:02:48 0g/s 68490p/s 68490c/s 68490C/s blkjandy..claciora
0g 0:00:02:49 0g/s 68488p/s 68488c/s 68488C/s sidux..jmtac
0g 0:00:02:50 0g/s 68487p/s 68487c/s 68487C/s 14872z..1304at
0g 0:00:02:51 0g/s 68482p/s 68482c/s 68482C/s muake7..mh166b
```

FIGURE 12 : THIS IS THE HARD PASSWORD, MIX OF NUMBER AND LETTERS, BRUTEFORCE, PASSWORD “QWEDF44535”


```
kali@kali: ~  
Session Actions Edit View Help  
fopen: /usr/share/wordlists/linuxwz.txt: No such file or directory  
  
(kali@kali)-[~]  
$ john --wordlist=/home/kali/Desktop/linuxwz.txt /home/kali/Desktop/hashz.txt  
Using default input encoding: UTF-8  
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])  
Cost 1 (HMAC size) is 0 for all loaded hashes  
Will run 2 OpenMP threads  
Press 'q' or Ctrl-C to abort, almost any other key for status  
0g 0:00:02:21 32.77% (ETA: 10:34:06) 0g/s 65879p/s 65879c/s 65879C/s 37C6TFhD  
8.. Anila1203  
0g 0:00:02:23 33.23% (ETA: 10:34:06) 0g/s 65856p/s 65856c/s 65856C/s Ed2keLDX  
.. 123451234  
0g 0:00:04:24 61.29% (ETA: 10:34:06) 0g/s 65825p/s 65825c/s 65825C/s kozaqupu  
.. chevysux77  
0g 0:00:04:26 61.75% (ETA: 10:34:06) 0g/s 65826p/s 65826c/s 65826C/s evw1qv..  
99979481  
0g 0:00:06:40 92.92% (ETA: 10:34:06) 0g/s 65864p/s 65864c/s 65864C/s mitwoch6  
31.. Coldplay15123  
0g 0:00:06:43 93.61% (ETA: 10:34:06) 0g/s 65869p/s 65869c/s 65869C/s matrix_2  
003.. bamibye  
0g 0:00:07:10 DONE (2026-03-14 10:34) 0g/s 65918p/s 65918c/s 65918C/s chiefie  
.. Egor2003  
Session completed.  
  
(kali@kali)-[~]  
$
```

FIGURE 13 : THIS IS THE EASY PASSWORD, MIX OF NUMBER AND LETTERS AND A SPECIAL CHARACTER AND A CAPITAL LETTER, WORDLIST, PASSWORD “PASS671!”

```
kali@kali: ~  
Session Actions Edit View Help  
.. chevysux77  
0g 0:00:04:26 61.75% (ETA: 10:34:06) 0g/s 65826p/s 65826c/s 65826C/s evw1qv..  
99979481  
0g 0:00:06:40 92.92% (ETA: 10:34:06) 0g/s 65864p/s 65864c/s 65864C/s mitwoch6  
31.. Coldplay15123  
0g 0:00:06:43 93.61% (ETA: 10:34:06) 0g/s 65869p/s 65869c/s 65869C/s matrix_2  
003.. bamibye  
0g 0:00:07:10 DONE (2026-03-14 10:34) 0g/s 65918p/s 65918c/s 65918C/s chiefie  
.. Egor2003  
Session completed.  
  
(kali@kali)-[~]  
$ john --incremental /home/kali/Desktop/hashz.txt  
Using default input encoding: UTF-8  
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])  
Cost 1 (HMAC size) is 0 for all loaded hashes  
Will run 2 OpenMP threads  
Press 'q' or Ctrl-C to abort, almost any other key for status  
0g 0:00:00:02 0g/s 52983p/s 52983c/s 52983C/s joybb1.. juigee  
0g 0:00:00:04 0g/s 58949p/s 58949c/s 58949C/s lay25.. saddd1  
0g 0:00:04:29 0g/s 66959p/s 66959c/s 66959C/s rylemms.. ryncser  
0g 0:00:04:30 0g/s 66964p/s 66964c/s 66964C/s tovrdr.. ruvms  
0g 0:00:04:31 0g/s 66964p/s 66964c/s 66964C/s 1qt63 ... 1qq942  
0g 0:00:04:35 0g/s 66985p/s 66985c/s 66985C/s bccjpb.. bcs1ed  
Session aborted  
  
(kali@kali)-[~]  
$
```

FIGURE 14: THIS IS THE EASY PASSWORD, MIX OF NUMBER AND LETTERS AND A SPECIAL CHARACTER AND A CAPITAL LETTER, BRUTEFORCE, PASSWORD “PASS671!”

```

kali@kali: ~
Session Actions Edit View Help
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:04 0.51% (ETA: 10:59:23) 0g/s 34049p/s 34049c/s 34049C/s zplpcF4V.
.90901
0g 0:00:03:17 13.81% (ETA: 11:09:58) 0g/s 19788p/s 19788c/s 19788C/s 1723861.
.Margarethe277
0g 0:00:03:22 14.01% (ETA: 11:10:13) 0g/s 19614p/s 19614c/s 19614C/s NBQASNkC
tk1o..LZDMSFRE
0g 0:00:03:50 14.86% (ETA: 11:11:59) 0g/s 18256p/s 18256c/s 18256C/s brisadel
acruz..mart-07
0g 0:00:05:12 17.89% (ETA: 11:15:15) 0g/s 16227p/s 16227c/s 16227C/s Christop
her.Toffee..Rdaudw7i9t
0g 0:00:05:18 18.01% (ETA: 11:15:37) 0g/s 16033p/s 16033c/s 16033C/s medicos7
3..casting2006
0g 0:00:07:34 22.50% (ETA: 11:19:50) 0g/s 14019p/s 14019c/s 14019C/s lul lake
..ber738n
0g 0:00:09:11 25.47% (ETA: 11:22:14) 0g/s 13098p/s 13098c/s 13098C/s 12345mh.
.Purchaser
0g 0:00:10:29 27.84% (ETA: 11:23:51) 0g/s 12542p/s 12542c/s 12542C/s Fluegel4
75..Passat664
0g 0:00:12:22 30.73% (ETA: 11:26:26) 0g/s 11740p/s 11740c/s 11740C/s Illegal2
001!!..Nora1Linda2
0g 0:00:13:04 31.35% (ETA: 11:27:52) 0g/s 11332p/s 11332c/s 11332C/s 301185ap
..scott_koeneman
Session aborted

(kali@kali)-[~]
$

```

FIGURE 15 : THIS IS THE HARD PASSWORD, MIX OF NUMBER AND LETTERS AND A SPECIAL CHARACTER AND A CAPITAL LETTER, WORDLIST, PASSWORD “QWEDF44535*”

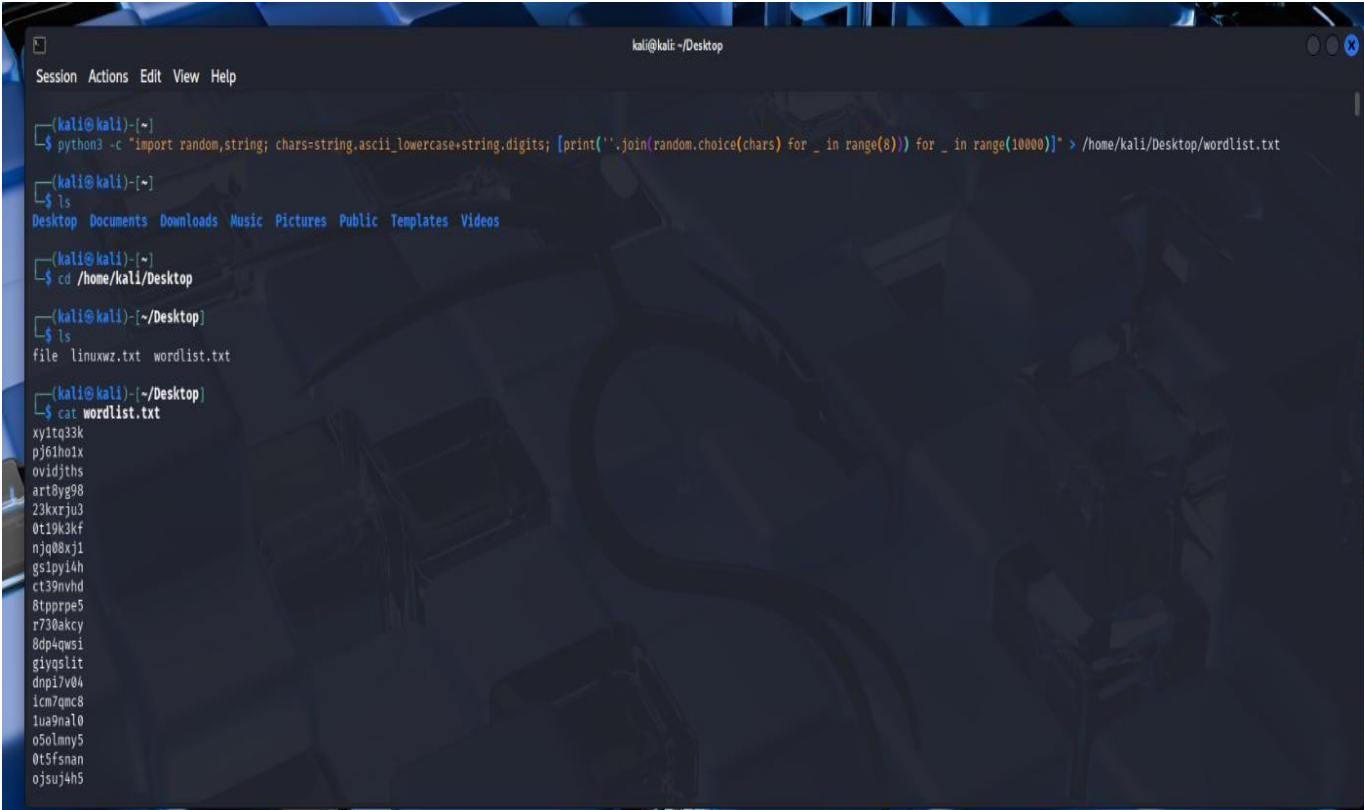
```

kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ john --incremental /home/kali/Desktop/hashz.txt
Using default input encoding: UTF-8
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])
Cost 1 (HMAC size) is 0 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:01 0g/s 42505p/s 42505c/s 42505C/s creal..breash
0g 0:00:01:31 0g/s 64721p/s 64721c/s 64721C/s thier5..thruam
0g 0:00:01:32 0g/s 64743p/s 64743c/s 64743C/s twhesi..tyruj5
0g 0:00:01:33 0g/s 64766p/s 64766c/s 64766C/s ttef01..ttbit0
0g 0:00:01:34 0g/s 64774p/s 64774c/s 64774C/s studomi..senaly3
0g 0:00:02:58 0g/s 64961p/s 64961c/s 64961C/s 145t1..mejhn
0g 0:00:03:45 0g/s 65223p/s 65223c/s 65223C/s rooc2c..royss3
0g 0:00:03:46 0g/s 65234p/s 65234c/s 65234C/s renark..regukz
0g 0:00:03:51 0g/s 65276p/s 65276c/s 65276C/s rmcced..rms239
0g 0:00:03:52 0g/s 65283p/s 65283c/s 65283C/s rblu1!..rb35sc
0g 0:00:03:53 0g/s 65290p/s 65290c/s 65290C/s rc257a..rcaatt
0g 0:00:03:58 0g/s 65301p/s 65301c/s 65301C/s jjr05j..jr1tyu
Session aborted

```

FIGURE 16:THIS IS THE HARD PASSWORD, MIX OF NUMBER AND LETTERS AND A SPECIAL CHARACTER AND A CAPITAL LETTER, BRUTEFORCE, PASSWORD “QWEDF44535*”

➤ Appendix B : Advanced Attack Scenario (Custom Wordlist and Caesar Cipher)



```
kali@kali: ~/Desktop
Session Actions Edit View Help

(kali@kali)~$ python3 -c "import random,string; chars=string.ascii_lowercase+string.digits; [print(''.join(random.choice(chars) for _ in range(8))) for _ in range(10000)]" > /home/kali/Desktop/wordlist.txt

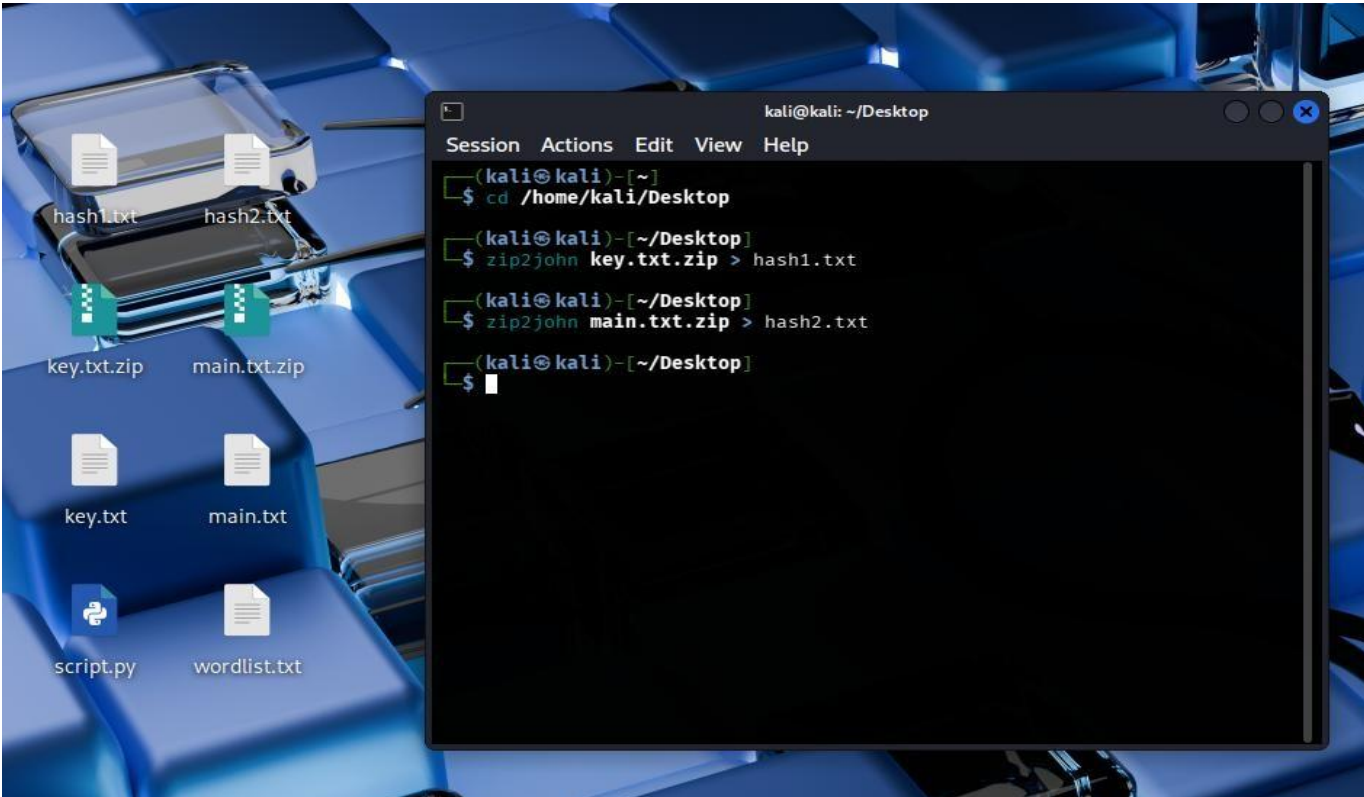
(kali@kali)~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos

(kali@kali)~$ cd /home/kali/Desktop

(kali@kali)~/Desktop$ ls
file linuxwz.txt wordlist.txt

(kali@kali)~/Desktop$ cat wordlist.txt
xy1tq33k
pj61hoix
ovidjths
art8yg98
23kxrju3
0t19k3kf
njq08xj1
gs1py14h
ct39nvhd
8tporpe5
r730akcy
8dp4qws1
giyqslit
dnp17v04
icm7qmc8
lua9na10
o5olmny5
0t5fsnan
ojsuj4h5
```

FIGURE 17 : OWN WORDLIST



```
kali@kali: ~/Desktop
Session Actions Edit View Help

(kali@kali)~$ cd /home/kali/Desktop

(kali@kali)~/Desktop$ zip2john key.txt.zip > hash1.txt

(kali@kali)~/Desktop$ zip2john main.txt.zip > hash2.txt

(kali@kali)~/Desktop$
```

FIGURE 18 :THE ORIGINAL FILES, THEIR ZIPS AND THE HASHES EXTRACTED FROM THOSE ZIPS USING JOHN

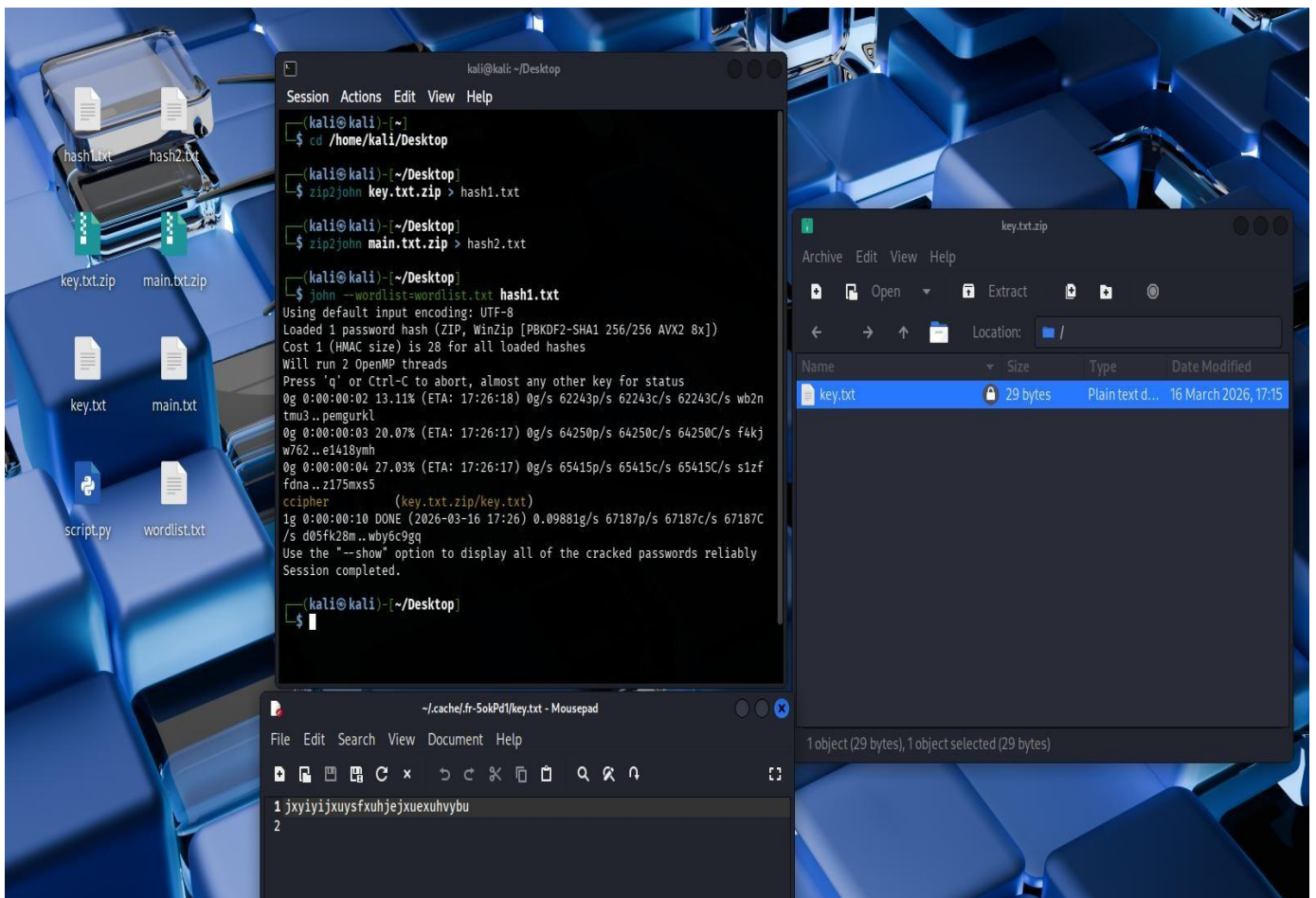


FIGURE 19 : HERE I HAVE OUR GENERATED WORDLIST FINDING THE PASSWORD FOR HASH FILE FOR ZIP. AS YOU CAN SEE THE TXT FILE CONTAINED THE CIPHER WHICH I WILL HAVE TO DECODE

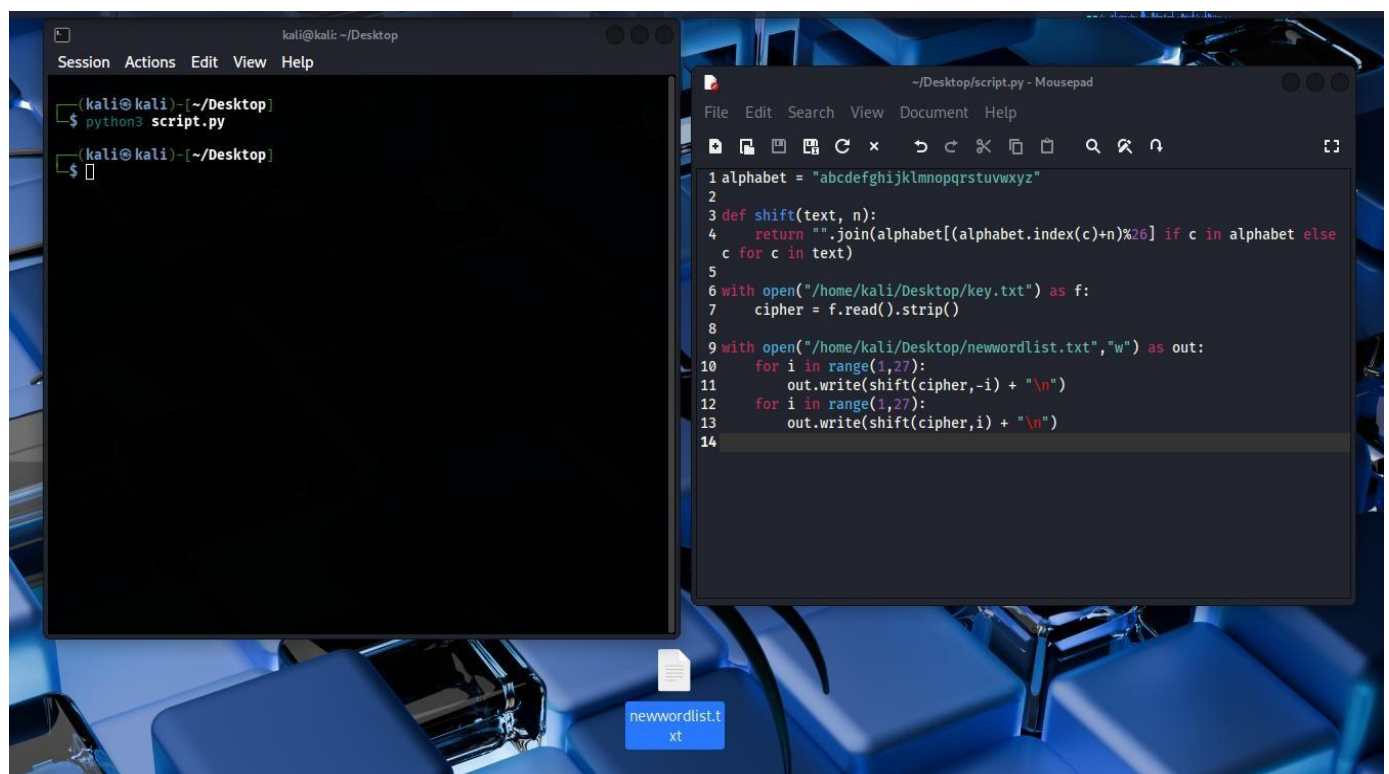


FIGURE 20 : ABOVE IS THE PYTHON SCRIPT WHICH , ONE AT A TIME, SHIFTS 26 LETTERS UP AND DOWN GIVING US A WORDLIST I CAN USE FOR MAIN.ZIP

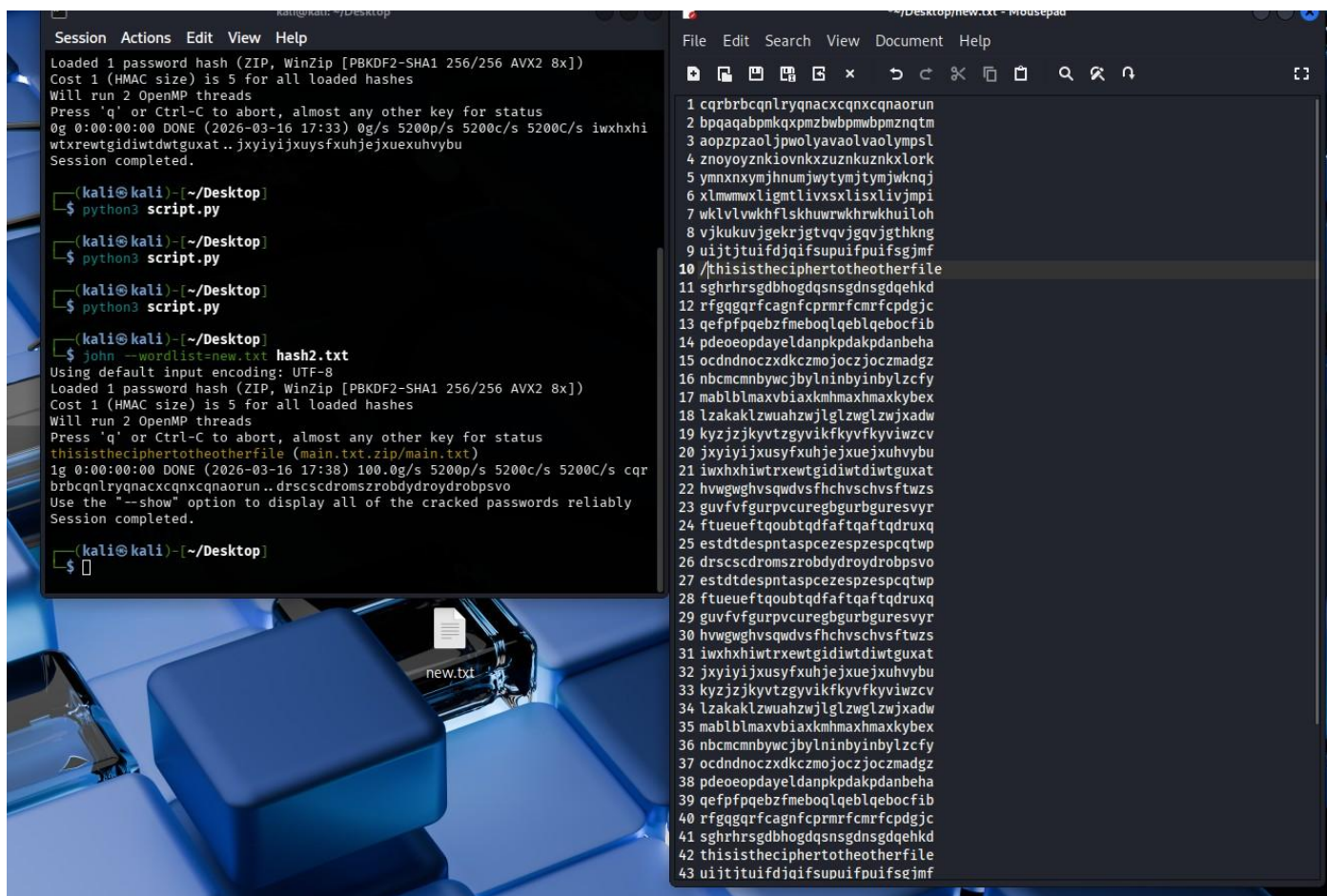


FIGURE 21 : HERE THE NEW WORD FILE EASILY FINDS THE ACTUAL PASSWORD

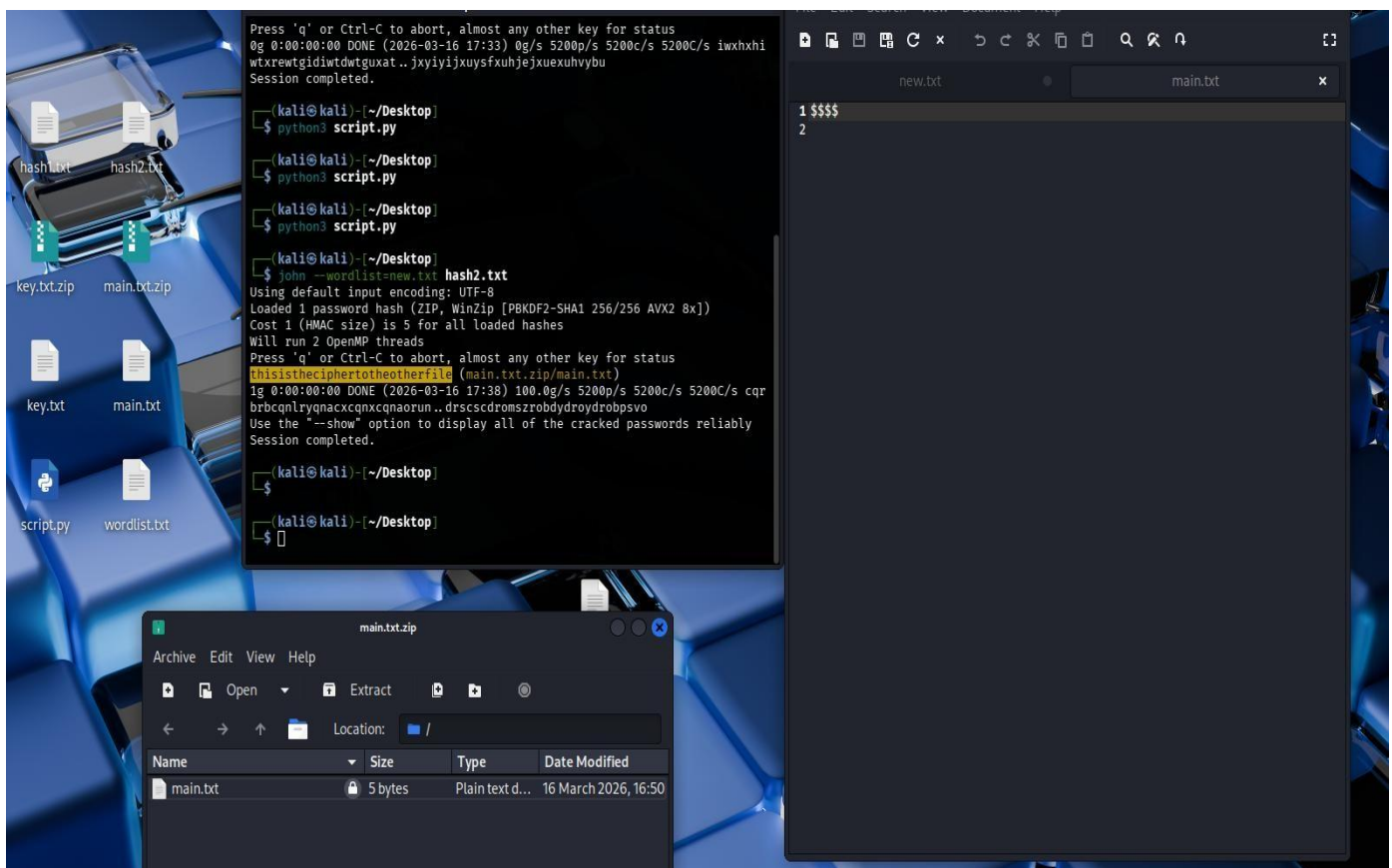


FIGURE 22 : THIS JUST SHOWS THE CONTENTS OF THE MAIN.ZIP FILE